

Abhishek Singh

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Profile

Final-year IIT Patna undergraduate with experience in Python, machine learning, stochastic simulation and model evaluation. Seeking to apply analytical modelling and AI to business decision-making as a Decision Analytics Associate.

Education

Indian Institute of Technology Patna 2023–2027 (expected)
B.Tech in Electronics and Communication Engineering CPI: 7.00/10
Completed coursework: Introduction to AI–ML; Random Signals and Stochastic Processes; Programming and Data Structures; Digital Signal Processing; Signals and Systems.
Ongoing coursework: Video Surveillance; Digital Image Processing.

Technical Skills

Programming: Python, C++, C
AI and data: NumPy, Pandas, PyTorch, TensorFlow, OpenCV; CNNs and autoencoders
Analytical methods: Predictive modelling, ROC-AUC evaluation, reconstruction-error analysis, stochastic simulation, parameter sweeps and reproducibility checks
AI platforms and tools: Hugging Face, Groq API, Llama 3; Git, Linux, AWS EC2

Research & Analytics Experience

Video Surveillance Research Intern | IIT Patna May–July 2026
Under Prof. Mahesh Kumar Kolekar Project repository

- Developed a PyTorch 3D convolutional autoencoder pipeline for video anomaly detection, covering preprocessing, training, inference and evaluation.
- Designed Hybrid Patch Feature Scoring combining patch reconstruction and latent-feature errors; achieved ROC-AUC of **0.734 on UCSD Ped2**, compared with a baseline of approximately 0.66–0.675.
- Compared anomaly scores and documented results through ROC curves and training-loss plots to assess model performance and communicate findings.
- Prototyped gradient-based anomaly localization and Llama 3 explanations through the Groq API to support interpretation of detected events.

Prabha / PEMAAN | Simulation & Model Validation | IIT Patna May 2026–Present
Photonic neural computing research under Prof. Sumanta Gupta

- Developed modular Python models of continuous-wave lasers and signed-weight photonic multiply–accumulate operations, including intensity noise and phase diffusion.
- Ran **1,000 stochastic realizations** and recovered a modelled 1 MHz laser linewidth within approximately **5.1%**; compared simulation estimates against theoretical predictions.
- Conducted power, wavelength, sampling-rate and random-seed validation; verified ideal multiply–accumulate outputs across three optical power levels.

Leadership & Communication

Software Head | Inditronix AI Labs April 2025–May 2026

- Led software architecture and coordinated a multidisciplinary team working on healthcare AI solutions; deployed infrastructure on AWS EC2 and developed ESP32-based IoT applications.

Creator | IITaku – Gyan, Code and Anime YouTube: 8,000+ subscribers

- Create educational content on programming and technical learning, communicating concepts to an online audience.

Achievements & Activities

- GATE ECE 2026:** AIR 1046; score 662. **JEE Advanced:** AIR 5701; **JEE Main:** AIR 4311.
- Kaggle:** Participated in healthcare, loan-payback and rainfall prediction competitions.
- Competitive programming:** Codeforces Pupil (peak 1241); CodeChef 3-star (peak 1643), best Div. 3 rank 150 and Div. 2 rank 459.